Wheat Seeds Dataset

Classification
&
Clustering

Wheat Seeds Dataset: Overview

- It is a dataset that contains measurements of the geometrical properties of kernels belonging to three different varieties of wheat.
- It was created in 2012 by researchers at the Institute of Agrophysics of the Polish Academy of Sciences in Lublin, Poland. Later, donated to the UCI Machine Learning Repository on September 28, 2012.
- Those researchers are:
  - Magorzata Charytanowicz
  - Jerzy Niewczas
  - Piotr Kulczycki
  - Piotr Kowalski
  - Szymon Lukasik

Wheat Seeds Dataset: Features

It contains seven features that can be used to describe the geometrical properties of wheat kernels. These features are:

1. **Area (mm^2)**: The total amount of space that it occupies.
2. **Perimeter (mm)**: The distance around it.
3. **Compactness**: A measure of how round a kernel is. A compact kernel is one that is close to being a circle.
4. **Kernel Length (mm)**: The longest dimension of the kernel.
5. **Kernel Width (mm)**: The shortest dimension of the kernel.
6. **Asymmetry Coefficient**: A measure of how asymmetrical a kernel is.
7. **Kernel Groove Length (mm)**: The total length of all the grooves on the kernel. The grooves are the small indentations that are often found on wheat kernels.

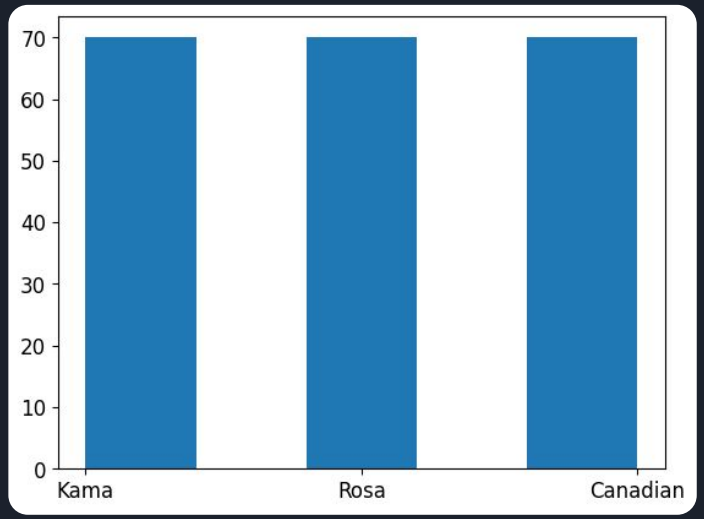
From now on the feature names will be referred by their original names (English)

Wheat Seeds Dataset: Classes

The dataset contains three different classes of wheat:

1. Kama
2. Rosa
3. Canadian

The dataset contains 70 samples of each of the three wheat varieties. The last column in the dataset contains class numbers (shown above) in which each of the sample belongs.

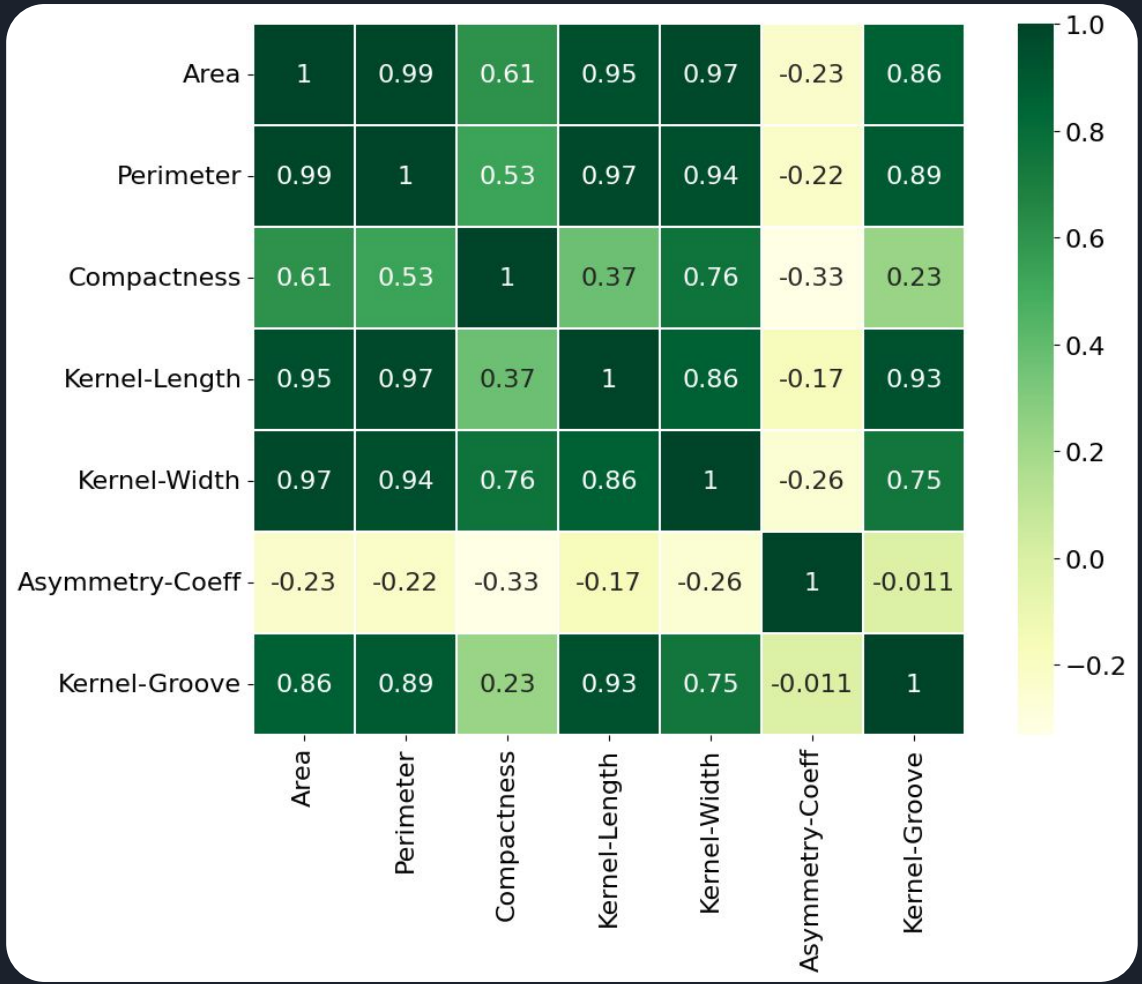


Wheat Seeds Dataset: Descriptive Statistics

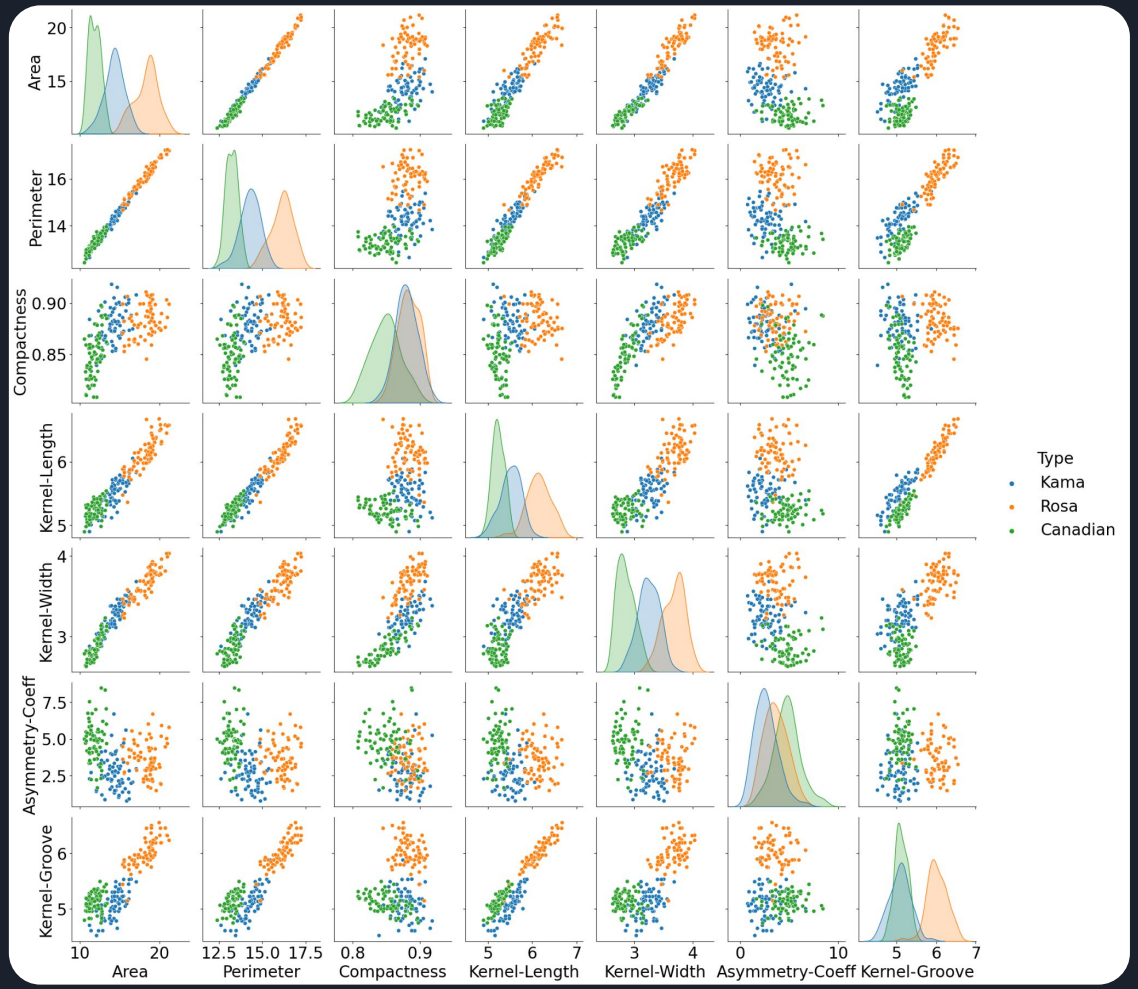
	Area	Perimeter	Compactness	Kernel Length	Kernel Width	Asymmetry Coeff.	Kernel Groove
Mean	14.84	14.55	0.87	5.62	3.25	3.70	5.40
Std.	2.90	1.30	0.02	0.44	0.37	1.50	0.49
Min	10.59	12.41	0.80	4.89	2.63	0.76	4.51
Max	21.18	17.25	0.91	6.67	4.03	8.45	6.55

There are 210 samples in the dataset and no missing values.

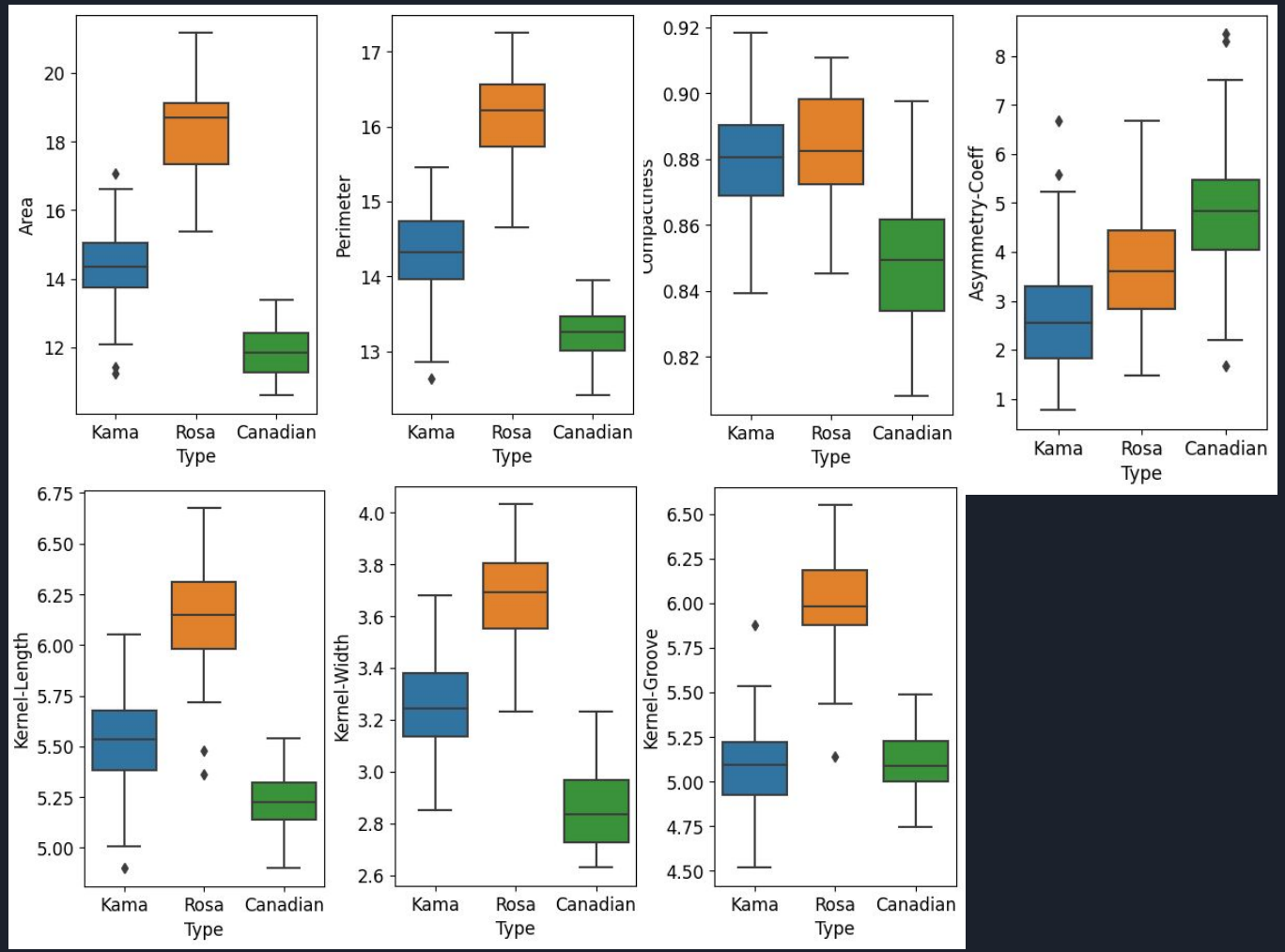
Features Pairwise Correlation

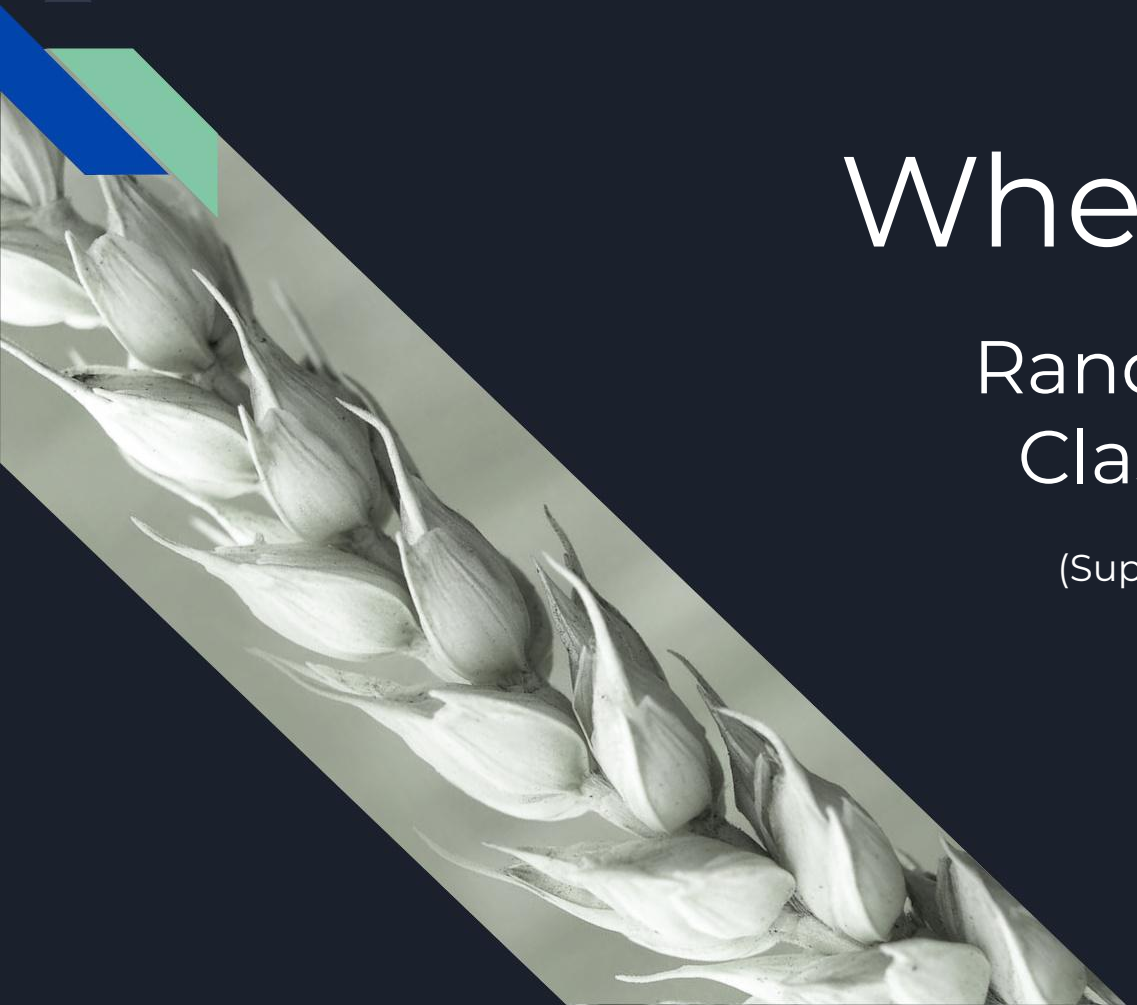


Features Pairplot



Features vs. Classes Boxplot





Wheat Seeds

Random Forest Classification

(Supervised Learning)

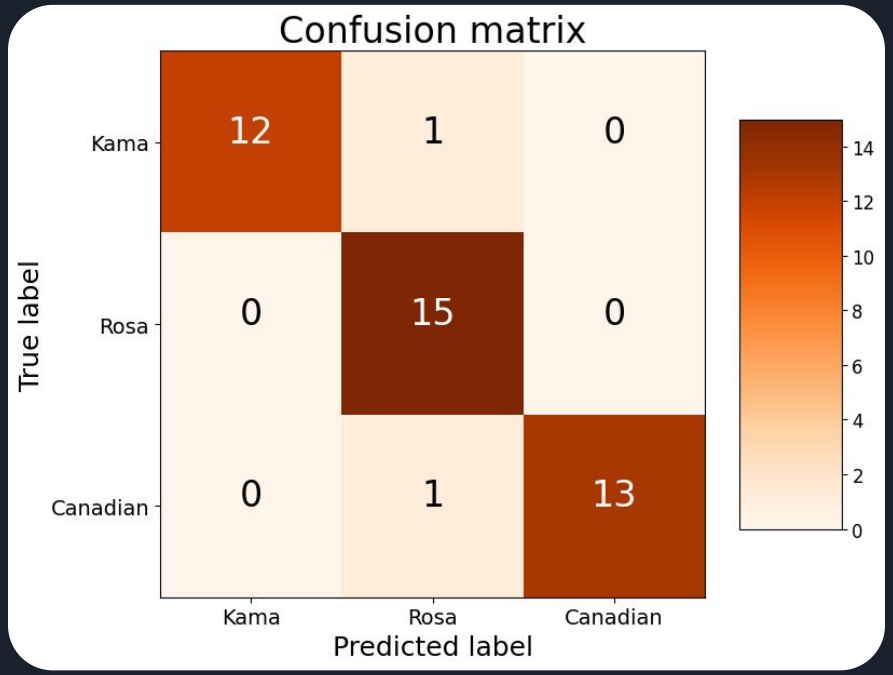
Random Forest Classification: Overview

- Random Forest is an ensemble learning method in machine learning.
  - Meaning, it uses multiple decision trees in training and prediction.
- Decision Tree is used in supervised learning for both classification and regression tasks.
  - It builds a tree-like structure to make decisions or predictions based on input data.
  - They are intuitive and interpretable, making them valuable tools for a variety of applications.

Random Forest helps address some of the limitations of single decision trees, such as overfitting. And it will be chosen based on these advantages.

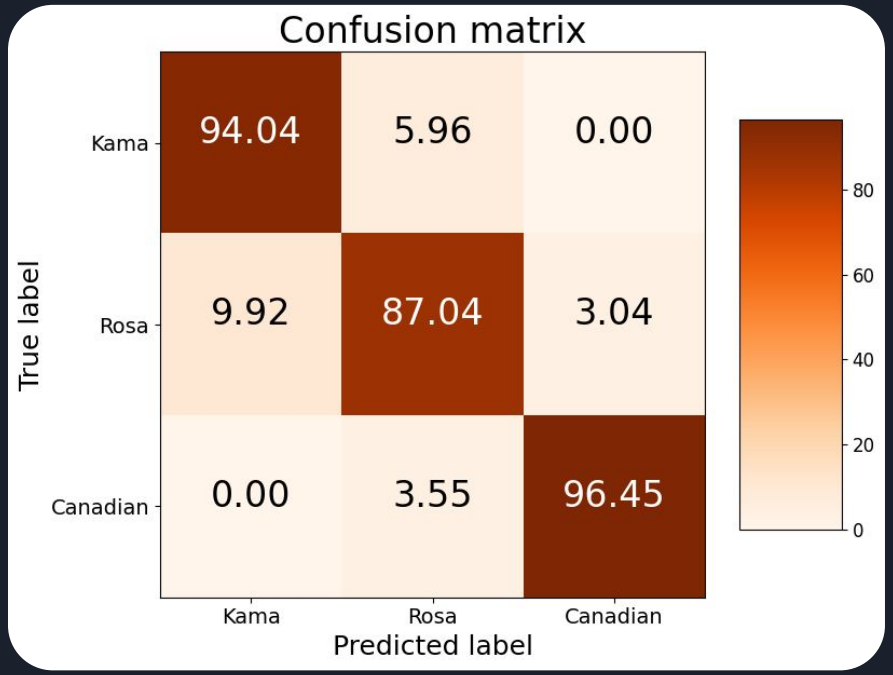
Random Forest Classification: Confusion Matrix

- Algorithm used:
  - RandomForestClassifier
- Input (X):
  - All seven features
- Target (y):
  - The three types of wheat
- Train-test split:
  - Train=80% (168 samples)
  - Test=20% (42 samples)
- Parameters: (Default)
  - n_estimators=100
  - criterion=gini
- Prediction Accuracy: 95.24%



Random Forest Classification: Confusion Matrix

- Averaged over a thousand splits
- Same:
  - Algorithm
  - Input (X)
  - Train-test split ratio
  - Parameters
- Normalized row
- Prediction Accuracy: 92.5%



Wheat Seeds

Hierarchical Clustering

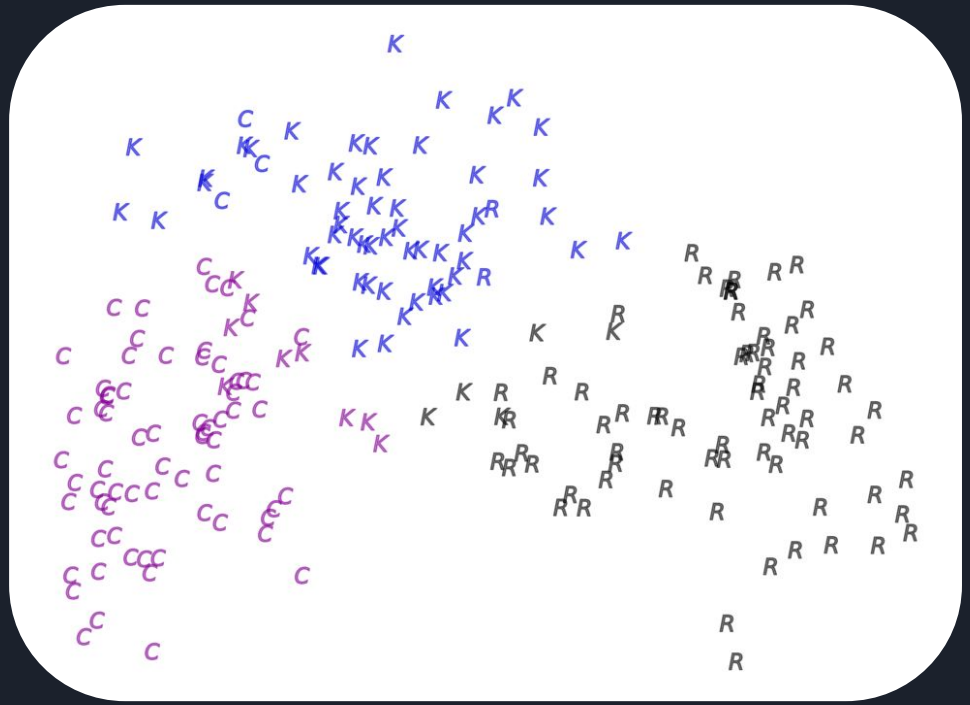
(Unsupervised Learning)

Hierarchical Clustering: Overview

- Hierarchical clustering is a popular technique used in cluster analysis, which is a type of unsupervised machine learning
- It is used for grouping similar objects or data points into clusters based on their similarities
- Hierarchical clustering builds a hierarchy of clusters, also known as a dendrogram, where each data point is treated as a single cluster initially
- The algorithm then iteratively merges or divides these clusters until a hierarchical tree-like structure is formed.

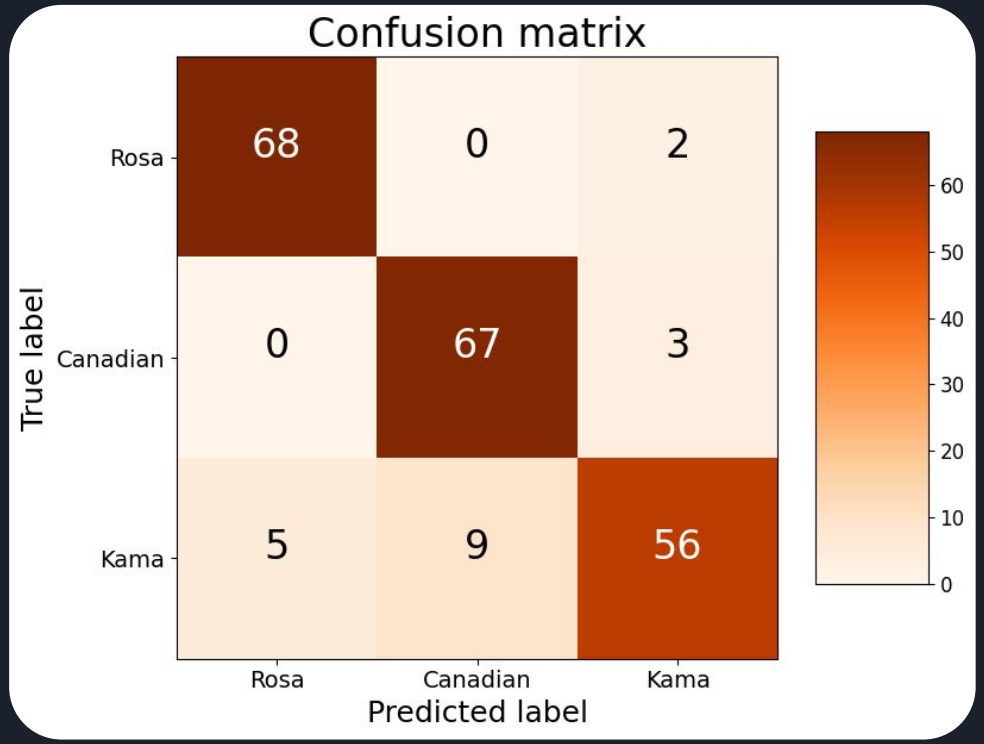
Hierarchical Clustering: Scatter Plot

- Algorithm used:
  - AgglomerativeClustering
- Input (210 samples):
  - Normalized
  - Selected two principal components using PCA from all seven features
- Parameters:
  - n_clusters=3
  - linkage=ward (Default)
- Markers:
  - K=Kama
  - R=Rosa
  - C=Canadian



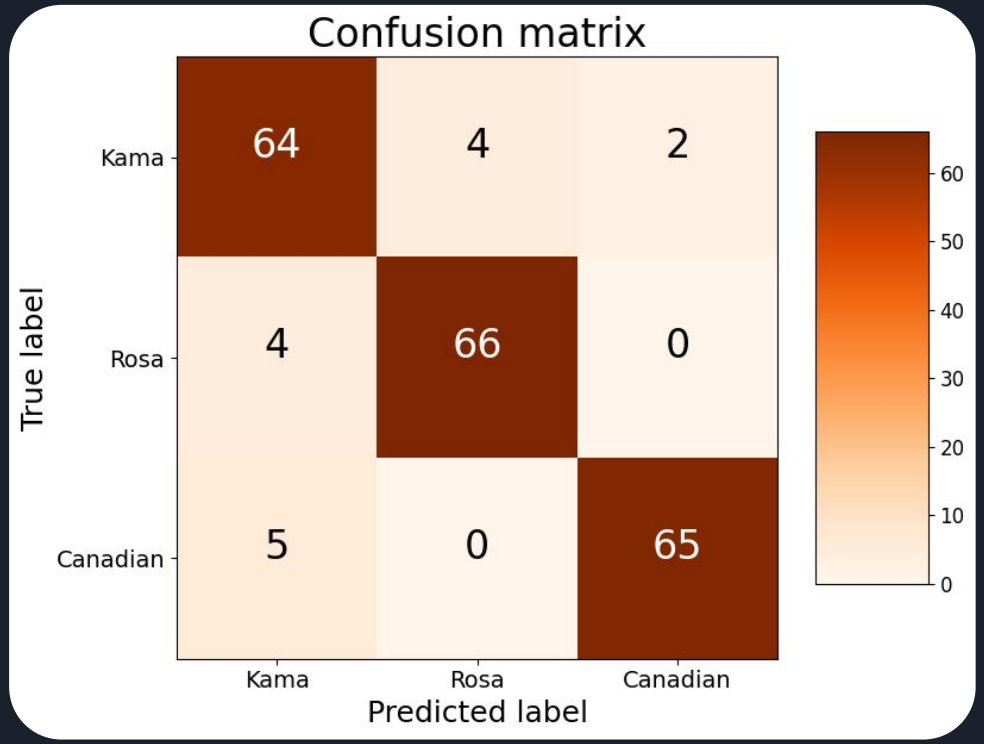
Hierarchical Clustering: Confusion Matrix

- Same:
  - Algorithm used
  - Input (210 samples)
  - Parameters
- Correctness:
  - Rosa: 97%
  - Canadian: 96%
  - Kama: 80%
  - **Total**: 91%



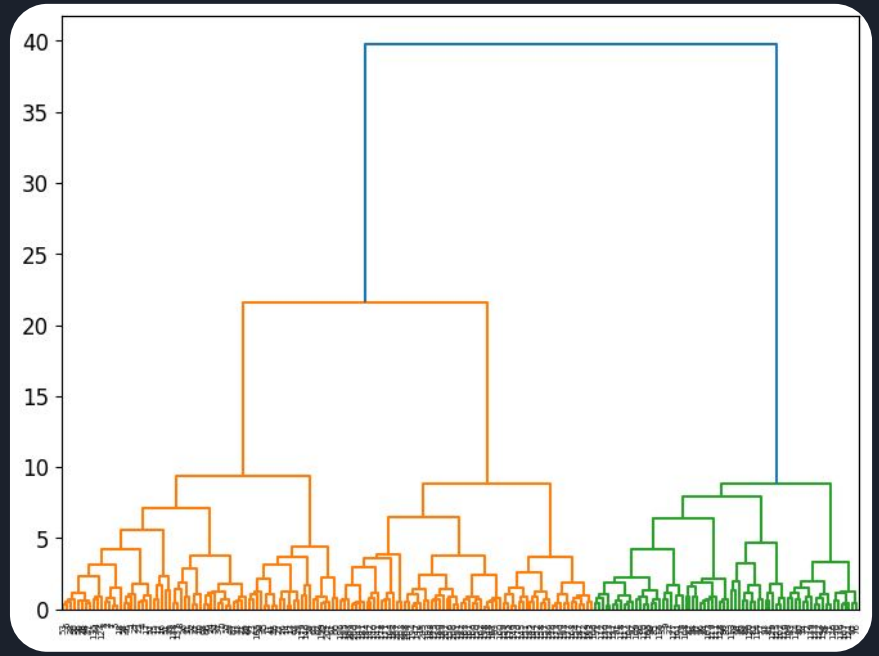
Hierarchical Clustering: Confusion Matrix

- Same:
  - Algorithm used
  - Parameters
- Input (210 samples):
  - All seven features
  - Normalized
- Correctness:
  - Kama: 91%
  - Rosa: 94%
  - Canadian: 93%
  - **Total**: 93%



Hierarchical Clustering: Dendrogram

- Algorithms used:
  - scipy.cluster.hierarchy.linkage
  - scipy.cluster.hierarchy.dendrogram
- Input (210 samples):
  - All seven features
  - Normalized
- Parameters:
  - linkage method=ward



End of Presentation

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